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MIS41130 Statistical Method Team Assignment

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Declaration of Authorship

We declare that all material in this assessment is our own work except where there is clear acknowledgement and appropriate reference to the work of others.

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Executive Summary

Maritime freight is central to Ireland's logistics network, and more than 90 percent of national trade in terms of volumes are carried via ports (CSO, 2024a). Based on the dataset retrieved from the Central Statistics Office (CSO) on the Tonnage of Goods Handled by Irish Ports 2020-2025 (quarterly), this paper evaluates the trends in port dominance, trade balance, seasonality, regional trade orientation, and resilience of Ireland's trade to significant shocks like COVID-19 and Brexit. Other CSO datasets that spanned pre-pandemic and pre-Brexit durations (2017 – 2019) were included to reinforce the analysis (CSO, 2024b).

The Descriptive analysis indicates that Dublin Port has dominated the Irish maritime freight with almost half of all tonnage in the country. Ireland remains regionally dependent as a significant portion of its trade is with the European Union and Great Britain remain. Although there is some descriptive evidence that suggests slightly seasonality due to higher tonnage in Q4, the difference between quarters is relatively small. These findings coupled with the fact that imports tonnage significantly outweighs the export tonnage supports the finding that Ireland's port system is vulnerable to geopolitical and regulatory shocks (Davies and Francois, 2021).

Inferential analysis were also conducted to evaluate Ireland's maritime activities. A two-sample t-test proves that imports are processed in considerably higher volumes than exports. An ANOVA test shows ports vary in tonnage handled and a regression model is used to determine that Dublin's maritime dominance is stable and not increasing over time. A paired t-test does not reveal any statistically significant difference in port activity following COVID-19, which is an indication of high operational recovery in line with the international results on maritime resilience (Notteboom, Pallis and Rodrigue, 2021). However, a different paired t-test gives a solid indication that the level of trade with Great Britain and Northern Ireland has decreased following Brexit.

Collectively, these findings suggest that the port network in Ireland is strong yet highly concentrated, import-based and geopolitically vulnerable. The results justify the suggestions of diversification of capacity, better resilience planning, enhanced direct EU trade routes, and specific assistance to exporters.

Introduction

The Central Statistics Office (CSO, 2024a) indicates that the significant portion of trade in Ireland is facilitated by ports, with more than 90 percent of all the trade activities in Ireland taking place through ports.

The analysis will be based on publicly available secondary data from the CSO on the Tonnage of Goods Handled by Irish Ports within the period between Q1 2020 and Q2 2025. The variables in the dataset will be port, direction of trade (received/forwarded), tonnage (000), quarter, year and region of trade. The variables allow to study the dominance of ports, the balance of imports and exports, seasonal changes, supply chain sustainability and regional trade patterns. The research aims to answer the following research questions:

1. Which port accounts for the largest share of national tonnage, and are differences across ports statistically significant?
2. Is Dublin's maritime dominance increasing over time?
3. Are imports handled in significantly different volumes than exports?
4. Does tonnage vary significantly across quarters (Q1–Q4), indicating seasonality?
5. How did trade volumes change during disruptions (COVID-19 and Brexit)?
6. By how much did trade with Great Britain change following BREXIT?

Overall, the purpose of the report is to examine the port-level goods handling in Ireland to determine patterns that can be used in planning capacity, operational resilience, and policy development using statistical methods.

Descriptive Statistics

This section summarizes Irish port activity and examines activity across ports, regions of trade, and direction of trade (export vs import) using the summary tables and visualizations constructed in Excel.

Table 1: Summary of Descriptive Statistics

Summary Table (000 Tonnes)	
Mean	209.03
Standard Error	11.32
Median	60
Mode	0
Standard Deviation	410.20
Sample Variance	168265.8
Kurtosis	11.70
Skewness	3.28
Range	2826
Minimum	0
Maximum	2826
Sum	274460
Count	1313
Quartile 1	6
Quartile 3	184

Measures of Central Tendency

Mean: The average tonnage (209,030 tonnes per quarter) is a very useful measure when comparing maritime activity across ports or regions or classes of trade. The mean is however sensitive to outliers, which are prevalent in maritime freight data. Large bulk shipments can particularly increase the average, especially in busy ports such as Dublin.

Median: The median of 60,000 tonnes is a more accurate measure of typical quarter to quarter port activity in this dataset. It is however significantly smaller than the mean, which indicates a right-skewed distribution.

Mode: The mode in this dataset is zero, indicating times of no activity in the ports, especially in smaller ports like Bantry bay, Waterford and Greenore. Even though mode is not as informative in numerical data, in this case, it emphasises the frequency of ports with little or no throughput.

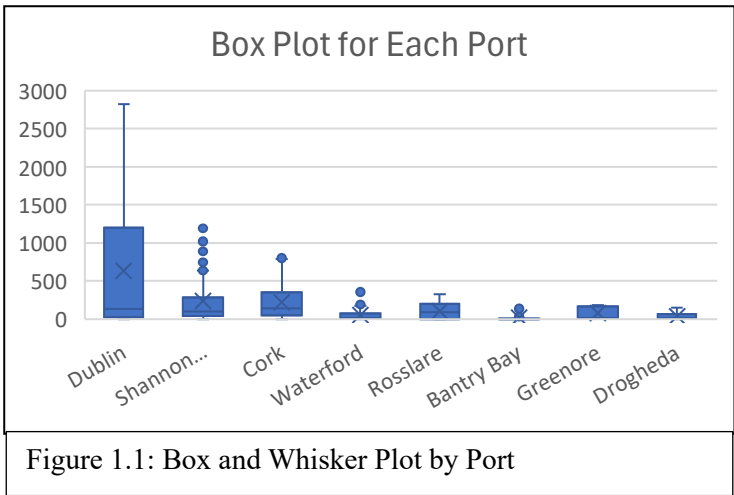
Measures of Dispersion

Standard Deviation and variance: The standard deviation of the maritime data in Ireland is 410,000 indicating that there are big variations in port activity that could be due to seasonal factors, irregular bulk shipments, or varying operational intensity among ports.

Range: The range of this case, 0 -2,826,000, is very high, which means that ports have freight movements of large magnitude and periods of low activity.

Interquartile Range (IQR): The IQR is 178,000, which means that the variability is significant with the 50 percent of the values spread widely between 6,000 and 184,000. A closer look at the Box and whisker plot for each Port in figure 1.1 supports this finding. With Dublin showing a significantly high Upper Bound when compared to the rest of the ports.

Shape of the Distribution:



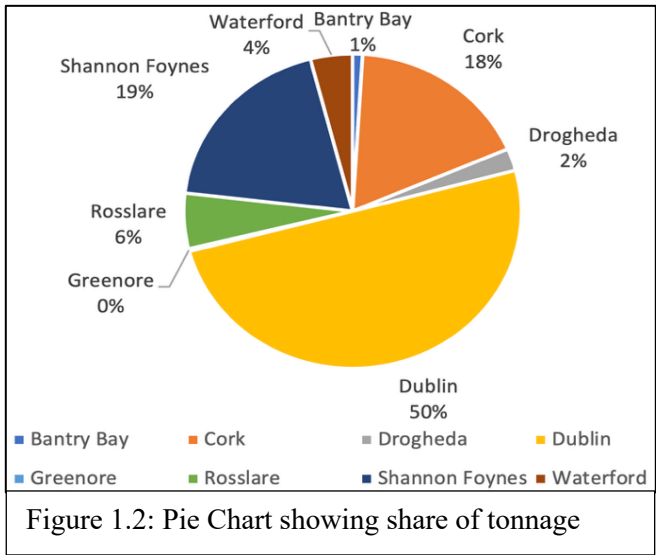
Skewness and Kurtosis:

Ireland’s port activity data shows a right skew of 3.28 which is because most ports and quarters exhibit moderate or low tonnage, while a few ports, such as Dublin have higher throughput. The kurtosis of 11.7 suggests that the data is heavily right tailed. This understanding provides the ground to test for what ports or factors are driving this skewness.

Key Takeaways from Descriptive Analysis:

Port-Level Concentration and Variability

Dublin dominates Ireland’s maritime freight sector, with a mean quarterly tonnage (630.54 thousand



tonnes) exceeding all other ports and accounts for 50% of all maritime tonnage in Ireland. The secondary level of major ports is Shannon Foynes and Cork, which are, however, at significantly lower levels. The high standard deviation of Dublin (782.07) shows that there is a lot of variation which is due to the high shipment peaks and the moderate throughput periods. This puts in perspective the possible capacity and resilience risks, since national trade flows depend on the operation of few large hubs.

Regional Dependence and Trade Orientation:

The descriptive analysis indicates that Ireland is highly oriented in terms of its trade to EU27 and Great Britain and Northern Ireland. The high dispersion of GB/Ni indicate that there is a lot of volatility.

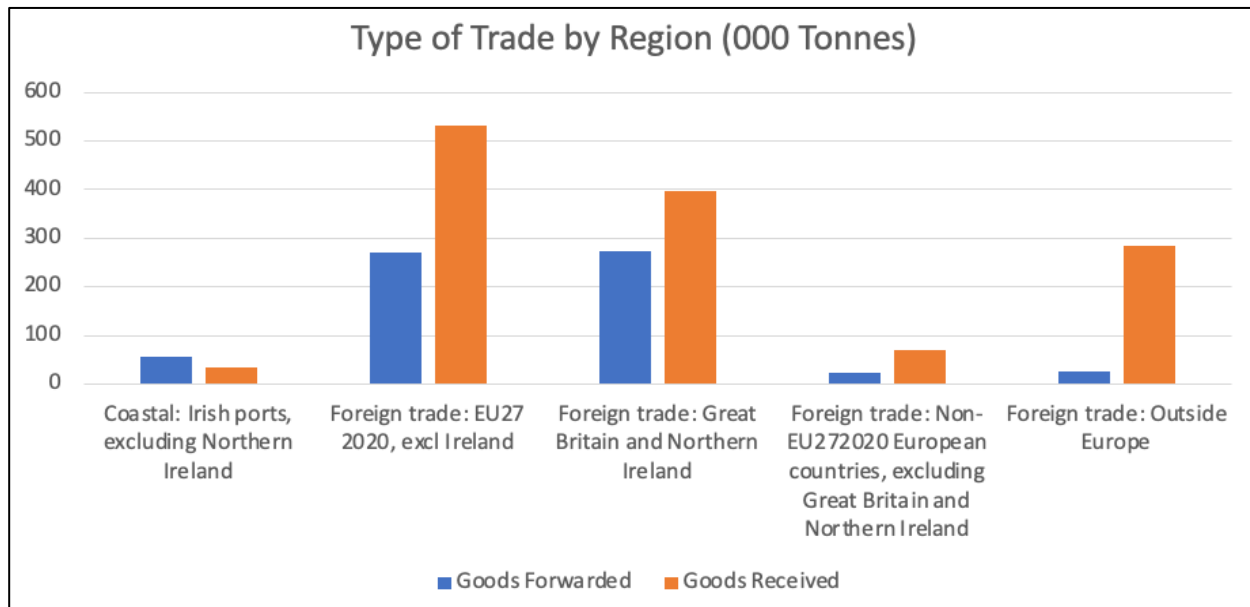


Figure 1.3: Bar Chart showing Trade Type by Region of Trade

Meanwhile, trade flows between Coastal Irish and non-European Europe are relatively small and stable. This concentration in trading partners underscores Ireland's vulnerability to regulatory and political developments within Europe (such as Brexit).

Import–Export Imbalance:

The average import is about two times higher than the average export (mean imports = 274.54, mean exports = 140.15). Both distributions are highly variable and skewed to the right, but imports are more widely distributed and have a higher number of peak volumes.

Seasonality in Port Activity:

The data shows a low to moderate seasonal trend with higher activity in Q4 (293.80 thousand tonnes). Q1-Q3 have similar mean tonnage. Consequently, an ANOVA test will be included to formally determine whether such patterns and seasonal trends are statistically significant.

Table 2: Quarterly mean Tonnage

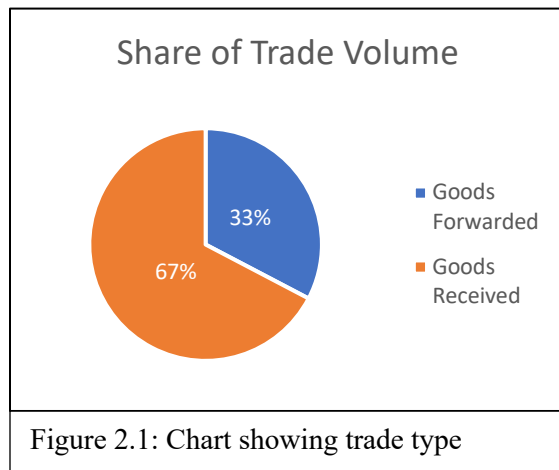
	Total (000)	Mean (000)
Quarter 1	148526	269.56
Quarter 2	150181	271.58
Quarter 3	123906	268.78
Quarter 4	126333	293.80

Inferential Statistics

This section addresses the research questions using inferential methods such as hypothesis testing and ANOVA to answer those questions.

Hypothesis Test for Differences Between Imports and Exports:

The descriptive analysis and pie chart shows that exports (goods forwarded) are handled in lower volumes



than imports (goods received) across Irish ports. To verify that a two-sample t-test is constructed, denotes the sample mean tonnage for goods forwarded (exports) and the sample mean for goods received (imports). Two independent samples of 31 observations are considered in the analysis. Since the ratio of the sample variances is not more than two, the population standard deviations are assumed to be equal and a pooled two-sample t-test is applied.

The hypotheses are:

- $H_0: \mu_1 = \mu_2$ (no difference in mean tonnage)
- $H_1: \mu_1 - \mu_2 < 0$ (exports are lower than imports)

Resulting t-statistic:

- $t = -3.36$ with $df = 60$ and a one-tailed p-value of $p = 0.001$
- Critical value = -1.67 and $\alpha = 0.05$:

Decision: Reject H_0

Interpretation:

At 5% significance level, the test provides sufficient evidence to conclude that imports are handled in significantly larger volumes than exports. This implies that:

- Irish supply chains are more likely to be affected by disruptions in international supply chains.
- Diversification of source of imports policies can be used to minimize exposure to external shocks.

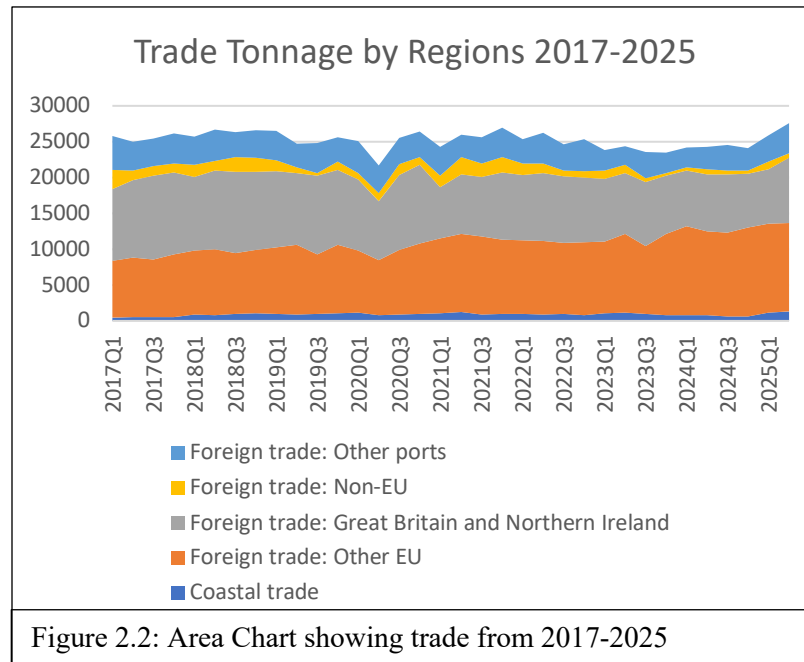
Recommendation:

- Invest more in berth capacity, storage and operational staffing to handle increased inbound volumes effectively.
- Develop and support special programs, including export incentives, efficient logistics support, and additional direct shipping routes, to contribute to the balancing of trade activity.

Hypothesis Test to Assess Ireland's Trade Resilience Around Disruptions

Pre-COVID vs Post-COVID

This research is to determine the extent to which the port activity in Ireland has declined during the post-



COVID-19 pandemic. As it can be seen in the area chart in figure 2.2, there is an indication that average trade tonnage indeed declined slightly after the pandemic. A paired t-test is performed comparing average quarterly tonnage in a pre-COVID (μ_1) baseline period (2017–2019) with tonnage in the post-COVID (μ_2) period (2022–2024). A paired sample of 31 quarterly observations has been collected to test this hypothesis.

The hypotheses are:

- $H_0: \mu_1 = \mu_2$ (no difference in mean tonnage between pre-COVID baseline and post-COVID levels)
- $H_1: \mu_1 - \mu_2 > 0$ (Pre-COVID averages was greater than post-COVID average)

Test Result:

- $t = 0.05$, $df = 30$ and p-value of $p = 0.48$.
- Critical value = 1.70 and $\alpha = 0.05$

Decision: Fail to Reject H_0

Interpretation:

At the 5% level of significance, the test does not provide any statistically significance evidence that pre-covid mean quarterly tonnage is higher than the post-covid average tonnage. This suggests that:

- The port system was well adjusted to disruptions caused by COVID-19.
- The variations that are normal variability, rather than long-term decline.

Pre-BREXIT vs Post-BREXIT:

Great Britain and Northern Ireland have historically been one of Ireland's biggest trading partners. This study evaluates whether Ireland's trade with GB/NI decreased in the period following Brexit. μ_1 denotes the quarterly mean tonnage traded with GB/NI prior to Brexit, and μ_2 denotes the quarterly mean tonnage post Brexit. The analysis considers a paired sample of 31 observations of quarterly GB/NI freight for both samples. Because the observations are not independent and form natural pairs, a paired-sample t-test is applied.

The hypotheses are:

- $H_0: \mu_1 = \mu_2$ (no change in mean quarterly tonnage with GB/NI before and after Brexit).
- $H_1: \mu_1 - \mu_2 > 0$ (the mean post-Brexit tonnage is lower than the mean pre-Brexit tonnage).

Resulting t-statistic:

- $t = 3.39, df = 30$ and p-value of $p = 0.001$.
- Critical value = 1.70 and $\alpha = 0.05$:

Decision: Reject H_0

Interpretation:

At the 5% significance level, the test provides sufficient evidence to conclude that the mean quarterly trade with Great Britain and Northern Ireland declined following BREXIT. This decline suggests:

- The emergence of new customs and regulatory obstacles probably decreased the effectiveness and the appeal of UK-related shipping routes
- Brexit-induced frictions have seen Ireland change its trade network to other partners, especially the EU27.

Recommendation to improve trade resilience:

- Invest in digital and resilient port operations.
- Improve contingency planning in the case of disruptions.
- Increase direct EU shipping links.
- Support companies changing to fit the UK trade

Confidence Interval of Change in Trade with Britain Following BREXIT

To confirm how much trade with Great Britain and Northern Ireland changed following Brexit a Confidence Interval is constructed at 95%.

Test Result:

The study finds that at 95% confidence interval; the mean tonnage traded with Great Britain and Ireland decreased by -1195.15 to - 245.04 thousand tonnes per quarter following BREXIT.

Interpretation:

This means that trade with Great Britain fell by 245 – 1195 on average following Brexit.

Analysis of Variance (ANOVA) Test for Seasonal Differences

This study aims to evaluate whether the mean tonnage handled varies consistently across the four calendar quarters of the year: Q1, Q2, Q3, and Q4. To investigate if there is any seasonality across the quarters, 31 observations have been collected for each of the four quarters. Because the factor of interest (Quarter) has more than two groups, and because each quarterly group represents an independent sample, a one-way ANOVA is applied. μ_1, μ_2, μ_3 & μ_4 denotes the population mean tonnage handled in Quarter 1, Quarter 2, Quarter 3, and Quarter 4, respectively.

As certain quarters (quarter 4) appear to show slightly higher average tonnage than others, ANOVA is used to assess whether these observed differences in sample means reflect true population differences.

The hypotheses are:

- $H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4$ (no difference in mean tonnage across quarters)
- $H_1: \mu_1 \neq \mu_2 \neq \mu_3 \neq \mu_4$ (at least one quarter exhibits a different mean tonnage level)

Test Result:

- $f = 0.08, df_N = 3, df_D = 169$, p-value of $p = 0.971$, critical value = 2.7 and $\alpha = 0.05$
- $SSTR = 64056919, SSE = 44844307003, MSTR = 21352306, MSE = 265350929$

Decision: Fail to reject the null hypothesis.

Interpretation:

At the 5% significance level, The ANOVA test provides insufficient evidence to conclude that the mean quarterly tonnage differs across Q1, Q2, Q3, and Q4. Although descriptive statistics suggested slightly higher averages in Q4, the extremely high p-value indicates that these differences are due to normal variation rather than true seasonal effects. This implies that:

- Ireland's port activity does not exhibit strong or consistent seasonality.
- Quarterly fluctuations observed in the descriptive charts reflect routine operational variability.

Recommendation:

- As port activity levels do not statistically differ by season, reducing the need for quarter-specific resource allocation or staffing.

Analysis of Variance (ANOVA) Test for Differences Between Ports

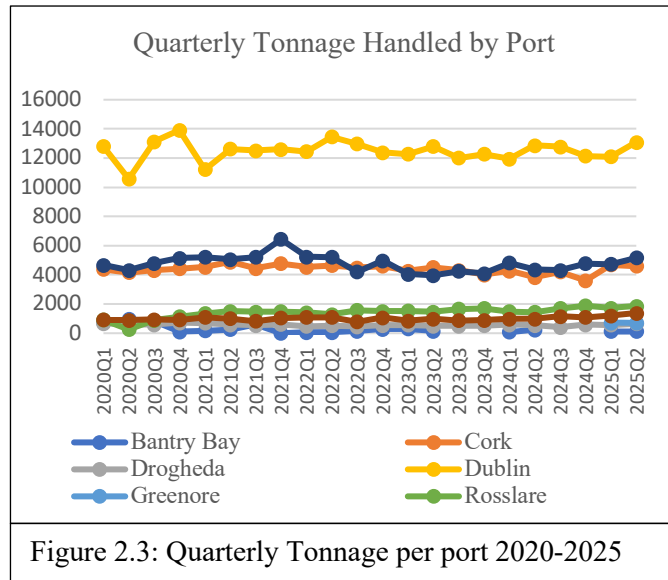


Figure 2.3: Quarterly Tonnage per port 2020-2025

The descriptive analysis and figure 2.3 shows that Dublin, on average, handles more tonnage than other ports. To test if this difference is statistically significant, this study considers 4 independent samples of 31 observations from Dublin, Cork, Shannon Foynes and Waterford (four of Ireland's largest ports). μ_1, μ_2, μ_3 and μ_4 are the mean quarterly tonnage handled at Dublin, Cork, Waterford and Shannon Foynes respectively. ANOVA is used to assess whether these observed differences in the descriptive analysis are statistically significant.

The hypotheses are:

- $H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4$ (no difference in mean quarterly tonnage between Ports)
- $H_1: \mu_1 \neq \mu_2 \neq \mu_3 \neq \mu_4$ (at least one Port handles a higher mean quarterly tonnage)

Test Result:

- $f = 16.36, df_N = 3, df_D = 116$, p-value of $p = 0.00$, critical value = 2.68 and $\alpha = 0.05$
- $SSTR = 16188371, SSE = 38251380, MSTR = 5396124, MSE = 329753$

Decision: Reject H_0

Interpretation:

At a 5% level of significance, the test shows strong statistical evidence, that the mean quarterly tonnage varies across ports. This confirms the huge position of Dublin in the maritime freight network of Ireland and implies that:

- Shocks to Dublin would have an excessively high effect on national trade flows.
- The port system in Ireland can be made more balanced in distribution of freight.

Recommendation:

Enhance and increase Shannon Foynes and Cork as a complementary gateway

Regression Model - Dublin Dominance Over Time

The ANOVA test confirmed Dublin as the dominant port in Ireland. However, to evaluate if Dublin’s dominance has been increasing over time, a regression model is applied.

Model Description:

The regression model examines whether Dublin’s share of national tonnage is growing, stable, or declining over time. The dataset consists of 260 observations across 34 quarters (2017Q1–2025Q2), comparing Dublin’s performance against all other ports combined.

Observed Market Share Trends:

Time Period	Dublin Share of Total
2017Q1	46.6%
2019Q2	53.7%
2021Q4	46.8%
2024Q2	53.0%

- Early period average (Q1–10): 50.0%
- Late period average (Q25+): 50.6%
- Change over time: +0.7 percentage points

Key Coefficient: Dublin × Time

- If positive and significant, Dublin’s dominance is increasing.
- If not significant, Dublin’s share is stable.

Test Result:

- $R^2 = 0.6717$ (model explains 67.2% of the variation)
- $Dublin \times Time = 148.97, p = 0.259$

Interpretation:

The positive coefficient of *Dublin x Time* in the regression model suggests Dublin’s advantage increased very slightly over time. However, this increase is not statistically significant. This suggests that:

- Dublin’s dominance is stable rather than accelerating.
- Other ports are neither catching up nor losing ground significantly and national port concentration is stable.

Recommendations to Improve Long-Term Port Balance:

- Develop long-term capacity plans for secondary ports: Even though Dublin’s dominance is stable, diversifying capacity now can prevent future congestion and reduce concentration risks.
- Monitor market-share trends regularly: Continued data-driven oversight will help detect emerging concentration patterns early and guide proactive infrastructure strategy.

Conclusion

In this report, the ports system of Ireland was analysed to assess the trends in maritime activity, trade flows, and resilience. The descriptive analysis showed that there are clear structural patterns. Dublin is the center of national throughput, imports are several times more significant than exports, and Ireland is dependent on the EU27 and Great Britain & Northern Ireland for its trade (CSO, 2024a; CSO, 2024b). Although there is a certain amount of quarterly variation, formal ANOVA results indicate that there is no statistically significant seasonality in port activity.

The imports-exports t-test showed that Ireland had a high dependence on imports, which means that Ireland is vulnerable to global disruptions. The comparison between ports shows that there are statistically significant differences in the average port throughput, which supports the risk of the heavy concentration of freight flows in one big hub in Ireland, a problem that is also described by Merk (2013). The COVID-19 resiliency test demonstrated that the maritime industry in Ireland had recovered to pre-pandemic level without any long-term structural deterioration, which aligns with the world recovery trends discussed by Notteboom, Pallis and Rodrigue (2021). Conversely, Brexit negatively affected Ireland's trade with Great Britain and Northern Ireland, which supports the larger economic findings of post-Brexit trade fallouts identified by Davies and Francois (2021).

Recommendation:

These findings have significant policy and port strategy implications. Diversification of port capacity should become a priority in Ireland to lessen the systemic dependency on Dublin and enhance resilience to disruptions in the future. Export promotion programs might also be used to deal with the chronic imbalance in imports and exports. Further establishment of direct EU maritime links can help to overcome the negative impact of Brexit and eliminate dependence on traditional UK transit routes. Lastly, resilience monitoring by conducting regular evaluations is necessary to inform long-term planning and ensure stability.

Appendices:

Appendix A:

Table 3: Summary table of port-level data

Port	total	mean	median	mode	St.dev	min	max	Quartile 1	Quartile 3	IQR
Bantry Bay	2618	22.00	0	0	47.59	0	209	0	4	4
Cork	48241	221.29	143	3	222.11	0	891	50.25	348.75	298.50
Drogheda	6244	34.69	14	0	41.97	0	161	3	61.25	58.25
Dublin	137458	630.54	132	8	782.07	0	2826	26	1196.75	1170.75
Greenore	719	79.89	18	4	83.87	4	187	10	157	147
Rosslare	15630	101.49	93	0	107.87	0	326	0	197.75	197.75
Shannon Foynes	52513	238.70	103	33	291.87	0	1193	42.5	271.25	228.75
Waterford	11037	56.60	18	0	78.79	0	354	4	73	69

Appendix B:

Table 4: Summary table of Regional Trade partner data

Regions	total	mean	median	mode	Std.dev	min	max	Quartile 1	Quartile 3	IQR
Coastal Irish	10473	44.95	6	0	61.10	0	263	0	79	79
EU27	116543	403.26	256	0	446.49	0	2061	92	453	361
Great Britain & NI	95968	334.38	93	0	631.51	0	2826	23.50	178	154.5
Europe non-EU27	11987	47.76	18	0	84.14	0	637	3	53.5	50.5
Outside Europe	39489	156.08	26	0	265.27	0	1077	4	141	137

Appendix C:

Table 5: Summary table for imports vs exports

Trade Type	total	mean	median	mode	Std.dev	min	max	Quartile 1	Quartile 3	IQR
Import	184767	274.54	78	0	503.93	0	2826	8	231	223
Export	89693	140.15	49	0	262.96	0	1599	5	142	137

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Contributions:

- Inioluwa Makinde (25%):** Led the report writing and research.
- Levi Simon-Eboh (25%):** Contributed to the Inferential Statistics analysis.
- Satvik Kumar (25%):** Conducted the Descriptive analysis.
- Mustafa Mohamed (25%):** Handled Data Cleaning and Visualization.